Resolving lunar and solar tidal responses with an optical clock baseline

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Gravity changes the rate at which clocks tick, making optical clocks probes of Earth's changing gravitational potential.[1,2] Resolving nearby tidal signals is difficult when clock comparisons are interrupted and restart at unknown frequency levels. Here we measure and separate neighbouring semidiurnal tidal responses using clocks in coastal Shanghai and inland Wuhan. The changing lunar–solar phase provides a reference across 545.9 hours of observations spread over 353 days. The measured spectrum shows local peaks near the principal lunar and solar frequencies, separated by 0.78 µHz. Relative to the combined solid-Earth and ocean-loading prediction, response scales are 1.00 ± 0.13 for M_2+N_2 and 0.93 ± 0.36 for S_2+K_2 (one statistical standard error). Optical-clock baselines can therefore distinguish neighbouring tidal contributions across interruptions without transferring a common frequency reference level between observing segments.

Earth's gravity field changes as oceans, atmosphere and the solid Earth respond to forces from the Moon and Sun. These variations carry information about mass redistribution and deformation, and enter the height and time references used to compare measurements between locations.[3,4,5] Optical clocks make this field observable through time itself: clocks at different gravitational potentials accumulate time at different rates.[1,2] Phase-stabilized fibre links now connect optical clocks across hundreds of kilometres, bringing direct, repeated comparisons of potential differences within reach.[6,7]

Laboratory redshift tests and remote clock comparisons have established measurements of static potential differences.[8,9,10] Following the changing Earth also requires the clocks to distinguish physical signals that evolve over hours. Tides provide a predictable test of this capability. Their astronomical frequencies and phases are known, while their effects on clock rates include direct forcing, Earth's deformation and ocean loading.[3] Resolving these variations matters both for geophysical measurements and for the time-dependent redshift corrections used in coordinated clock comparisons and common time scales.[3,11,12]

The principal lunar and solar semidiurnal tides provide a demanding pair of nearby signals. M_2 repeats every 12.42 h and S_2 every 12 h, giving a frequency separation of only 0.78 µHz. Over a short comparison, their variations can look much alike. Interruptions add a further difficulty because a new run may start at an unknown frequency level. Yet the lunar peaks arrive about 50 minutes later in solar time each day. Observations on different dates therefore sample different combinations of lunar and solar tidal phase, even when their frequency levels cannot be directly compared.

The astronomical phase remains defined at every actual observation time, including after an interruption. Independent segment constants can accommodate unknown reference levels, while the changes within each segment constrain common tidal responses at their known phases. Here we apply this reference to the strontium–ytterbium comparison between coastal Shanghai and inland Wuhan. The coastal–inland geometry makes ocean loading an appreciable contribution to the predicted potential difference. We jointly estimate the lunar semidiurnal group M_2+N_2 and the

neighbouring S_2+K_2 group. K_2 contains both lunar and solar forcing; the approximately daily tides remain an auxiliary physical component. The measured spectrum and group responses test separation across the interrupted record.

A coastal–inland prediction

The strontium clock in Shanghai and ytterbium clock in Wuhan are 692 km apart, joined through Hefei by a 1,350-km phase-stabilized fibre link (Fig. 1a).[13] The summer observations overlap the absolute frequency-ratio measurement reported by Hou and colleagues; the present observable is the time-dependent tidal response. The clock systems and their performance are described in their published evaluations.[14,15] The link extends a transmission architecture previously demonstrated over 2,067 km of field-deployed fibre.[16] We define a positive frequency-ratio variation as Wuhan Yb running faster relative to Shanghai Sr. In the geodetic potential convention used here, the measured fractional variation is $y = -[\delta W_{\mathrm{Wuhan}} - \delta W_{\mathrm{Shanghai}}]/c^2$.

Ocean mass redistribution produces different gravitational responses at coastal Shanghai and inland Wuhan. Their differential tidal signal therefore includes an appreciable loading contribution alongside the solid-Earth tide. The solid response combines the direct astronomical potential, the Earth's induced potential and displacement of the clocks. Ocean loading likewise includes water attraction, induced potential and clock displacement. We combine an IERS-based solid-tide prediction with FES2022 ocean elevations and a loading kernel containing all three loading contributions (Methods).[17,18,19,20]

The predicted M_2 amplitudes are 3.43×10^{-18} for the solid contribution and 1.42×10^{-18} for loading. Their phases differ by 19.8°, giving a combined amplitude of 4.78×10^{-18} (Fig. 1b). Loading is about 30% of the combined M_2 amplitude. On the sampled record, after removing segment means, its weighted root-mean-square variation is 33.1% of the combined prediction. These ratios describe the model contribution; the measurement tests the combined potential.

The record spans September 2025 to August 2026, with 54 valid segments in eight observing periods (Fig. 1c). Its 3,299 ten-minute means contain 545.9 h of accepted nominal one-second measurements. Observations occur in eight calendar months; the annual span is not continuous operation. Actual UTC times and all gaps are retained. The June–August observations supply 273.9 h and sample a broad range of relative lunar–solar phase, while the earlier periods add different phases and operating conditions (Fig. 1d). Both the duration and placement of the observations determine what can be separated.

Jointly resolved semidiurnal responses

The reference levels of the 54 observing segments are left independent. The analysis retains the variation within each segment and uses its actual UTC times to place that variation at the correct astronomical phase. All segments can therefore constrain common physical responses without an artificial joining of their frequency levels. The constants represent unknown reference levels, not measured frequency jumps. The full record reveals recurring modulation when ordered by M_2 phase (Fig. 2).

Joint estimation gives $A_L = 1.00 \pm 0.13$ for M_2+N_2 and $A_S = 0.93 \pm 0.36$ for S_2+K_2 (Fig. 3a). Each semidiurnal group has its own response coefficient within the same physical model. One denotes the specified combined prediction. The coefficients are estimated from the original ten-minute means using the same noise weights throughout. The reported errors are the larger of residual-scaled least-squares and UTC-day-deletion standard errors; their components and conditional interval coverage are given in Methods. No empirical daily sinusoid is subtracted before this joint estimate.

After subtracting the fixed minor tidal remainder, the observations and unit lunar prediction undergo the same projection against segment constants and the other two groups. Twelve phase bins reveal the recurring modulation, with the uncertainty of the jointly estimated nuisance terms propagated into the error bars. The complete set of individual observations and the support of each bin are shown below the phase means.

Allowing a lunar component one quarter-cycle from the predicted phase gives in-phase and quadrature responses of 1.00 ± 0.13 and -0.10 ± 0.20 (Fig. 3b), consistent with the astronomical phase.

Spectral separation through the observation window

The measured spectrum provides a direct view of the two principal semidiurnal neighbourhoods (Fig. 3c). An equal-weight sinusoidal scan with 54 segment constants has local maxima at 22.361 and 23.147 µHz, close to the predicted M_2 and S_2 frequencies of 22.364 and 23.148 µHz. The prescribed principal frequencies are 0.78 µHz apart. The left axis expresses spectral density in gravitational-potential units and the right in fractional-frequency units; a fractional variation of 10^{-18} corresponds to about $0.09\,\mathrm{m}^2/\mathrm{s}^2$ of potential difference.

The observation gaps produce sidelobes around a sampled tidal line. Single-constituent injections through the actual sampling and scan give 8.03% M_2-to-S_2 cross-power and 8.15% in the reverse direction. Conditional scans retain a local feature near each frequency after projecting out the sine and cosine at the other (Extended Data Fig. 1). Together with the joint physical-template estimates, these features identify distinct responses in the neighbouring semidiurnal groups. The two groups' prescribed internal structure and the sampling-window response are treated explicitly in Methods.

Consistency across observing periods

Observations before and during summer both contribute to the lunar response. With the same segment model, the coefficient is 1.21 ± 0.23 before summer and 0.77 ± 0.18 in summer. Deleting each observing period in turn gives point estimates from 0.885 to 1.093 (Fig. 4a). These overlapping fits show each period's influence on the full-record estimate.

Changing only the grouping of the reference levels gives similar lunar coefficients (Fig. 4b). The available interval and correction records do not establish a calibrated transfer of zero across interruptions, motivating the separate segment constants

used here. The eight-period design gives 0.987 ± 0.123 on the same observations, physical templates and weights. Its residual structure and the full comparison of group coefficients are given in Methods and Extended Data Fig. 2.

Daily temperature changes and operating schedules motivate a 24-h sensitivity check. Adding one shared 24-h sine–cosine pair to the present model gives $A_L = 1.04 \pm 0.13$. The matched summer round-trip monitor also leaves the lunar estimate essentially unchanged, with a paired shift of 0.0061 ± 0.0063 over 246.6 h. The measurement support and full results of that control are shown in Extended Data Fig. 3. Free 12-h terms are confined to the sensitivity analysis in Methods.

The Shanghai–Wuhan comparison measures tidal variations of the gravitational potential and separates neighbouring semidiurnal responses in an interrupted record. The principal-frequency peaks, conditional spectra and jointly estimated group amplitudes connect the observed modulation to the specified astronomical signals.

The measurement uses two references with different roles: each observing segment has its own frequency level, whereas the tidal forcing retains a common astronomical phase. That phase links the variations recorded within separate runs and allows their contributions to be estimated together. The observation window supplies separation through its phase coverage as well as its duration. Scheduling comparisons at different lunar–solar phases is therefore part of the measurement design, alongside clock stability and accumulated observing time.

On this coastal–inland baseline, the resolved response tests a joint prediction with a substantial ocean-loading contribution. Separately determining that contribution will require observations that better distinguish its amplitude and phase from the solid-Earth response. For developing clock networks, the present result provides a route to measuring distinct tidal contributions during intermittent operation. Independent, contemporaneous European clock-network observations have tracked the overall tidal potential on different baselines.[21]

References

1. Ludlow, A. D., Boyd, M. M., Ye, J., Peik, E. & Schmidt, P. O. Optical atomic clocks. *Rev. Mod. Phys.* **87**, 637–701 (2015). https://doi.org/10.1103/revmodphys.87.637
2. Mehlstäubler, T. E., Grosche, G., Lisdat, C., Schmidt, P. O. & Denker, H. Atomic clocks for geodesy. *Rep. Prog. Phys.* **81**, 064401 (2018). https://doi.org/10.1088/1361-6633/aab409
3. Voigt, C., Denker, H. & Timmen, L. Time-variable gravity potential components for optical clock comparisons and the definition of international time scales. *Metrologia* **53**, 1365–1383 (2016). https://doi.org/10.1088/0026-1394/53/6/1365
4. Denker, H. et al. Geodetic methods to determine the relativistic redshift at the level of 10^{-18} in the context of international timescales: a review and practical results. *J. Geod.* **92**, 487–516 (2018). https://doi.org/10.1007/s00190-017-1075-1
5. Schröder, S., Stellmer, S. & Kusche, J. Potential and scientific requirements of optical clock networks for validating satellite-derived time-variable gravity data. *Geophys. J. Int.* **226**, 764–779 (2021). https://academic.oup.com/gji/article/226/2/764/6212226
6. Lisdat, C. et al. A clock network for geodesy and fundamental science. *Nat. Commun.* **7**, 12443 (2016). https://doi.org/10.1038/ncomms12443
7. Riehle, F. Optical clock networks. *Nat. Photon.* **11**, 25–31 (2017). https://doi.org/10.1038/nphoton.2016.235
8. Chou, C. W., Hume, D. B., Rosenband, T. & Wineland, D. J. Optical Clocks and Relativity. *Science* **329**, 1630–1633 (2010). https://doi.org/10.1126/science.1192720
9. McGrew, W. F. et al. Atomic clock performance enabling geodesy below the centimetre level. *Nature* **564**, 87–90 (2018). https://doi.org/10.1038/s41586-018-0738-2
10. Grotti, J. et al. Geodesy and metrology with a transportable optical clock. *Nat. Phys.* **14**, 437–441 (2018). https://doi.org/10.1038/s41567-017-0042-3
11. Dimarcq, N. et al. Roadmap towards the redefinition of the second. *Metrologia* **61**, 012001 (2024). https://doi.org/10.1088/1681-7575/ad17d2
12. Lindvall, T. et al. Coordinated international comparisons between optical clocks connected via fiber and satellite links. *Optica* **12**, 843–852 (2025). https://doi.org/10.1364/OPTICA.561754
13. Hou, L. et al. Yb/Sr optical clock comparison at 2.2×10^{-18} uncertainty over a 1350-km fiber link. *arXiv*, 2610.34250v1 (2026). https://arxiv.org/abs/2610.34250v1
14. Jia, Z.-P. et al. Improved systematic evaluation of a strontium optical clock with uncertainty below 1×10^{-18}. *Metrologia* **63**, 025002 (2026). https://doi.org/10.1088/1681-7575/ae449e
15. Zhu, Q. et al. Ytterbium lattice clock with systematic uncertainty of 1.3×10^{-18} and instability at the 10^{-19} level. *arXiv*, 2606.10514v2 (2026). https://arxiv.org/abs/2606.10514v2
16. Chen, F.-X. et al. 10^{-21}-level optical frequency dissemination over 2067 km of noise-loaded field-deployed fiber network. *Light Sci. Appl.* **15**, 276 (2026). https://doi.org/10.1038/s41377-026-02299-1
17. Petit, G. & Luzum, B. (eds) IERS Conventions (2010). *IERS Technical Note* **36** (2010). https://iers-conventions.obspm.fr/content/tn36.pdf
18. CNES. FES2022 (Finite Element Solution) Tidal model (Version 2024). CNES https://doi.org/10.24400/527896/A01-2024.004 (2024).
19. Farrell, W. E. Deformation of the Earth by surface loads. *Rev. Geophys.* **10**, 761–797 (1972). https://doi.org/10.1029/RG010i003p00761
20. Han, D. & Wahr, J. The viscoelastic relaxation of a realistically stratified earth, and a further analysis of postglacial rebound. *Geophys. J. Int.* **120**, 287–311 (1995). https://doi.org/10.1111/j.1365-246x.1995.tb01819.x
21. Stahl, K. et al. Optical clock network tracks tidal variations of the gravity potential. *arXiv*, 2610.39635v1 (2026). https://arxiv.org/abs/2610.39635v1

Figure legends

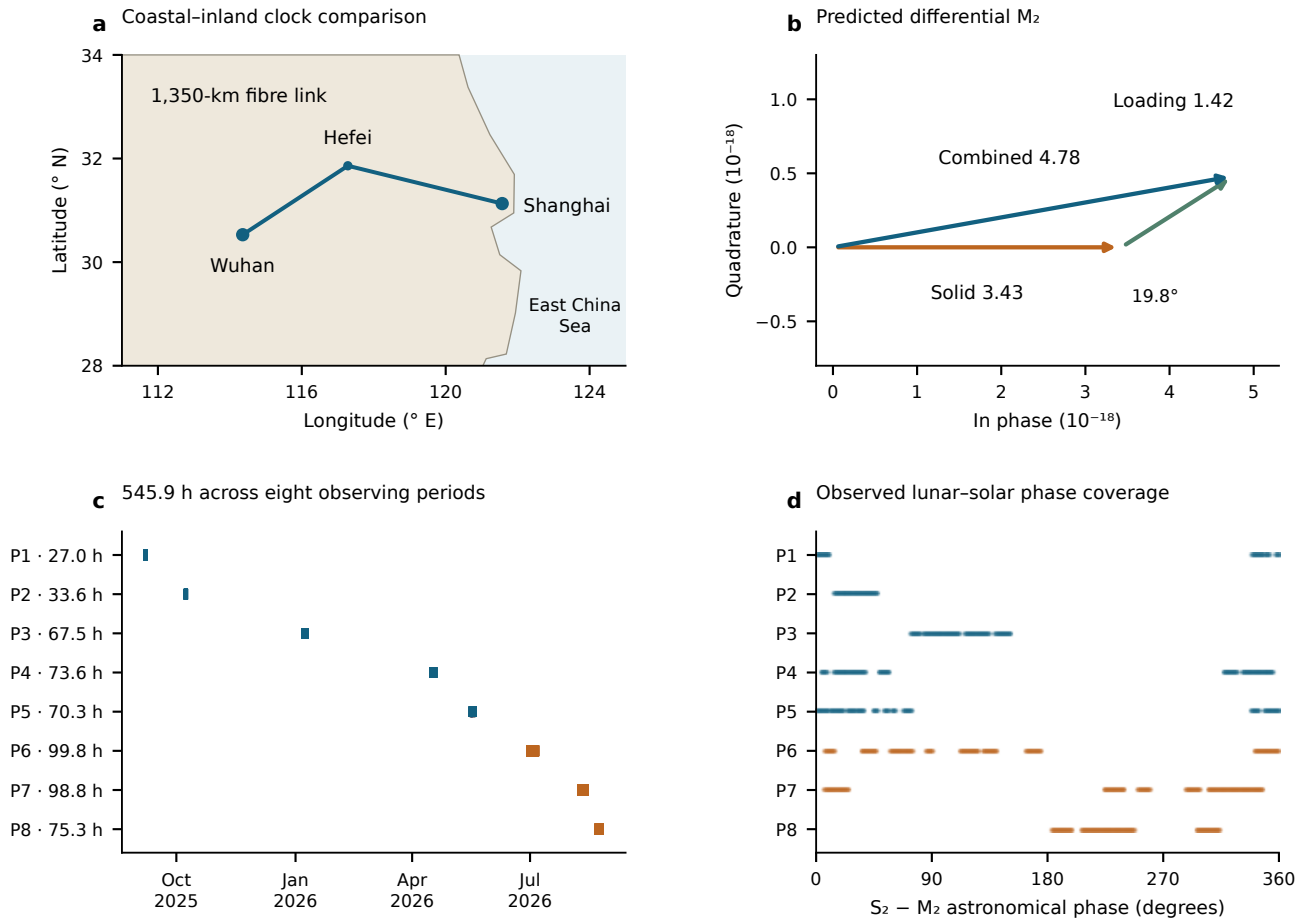


Fig. 1 | Geometry and changing lunar–solar phase. a, Shanghai Sr and Wuhan Yb, 692 km apart, with the 1,350-km fibre connection through Hefei drawn schematically. **b,** Predicted differential M_2 solid and ocean-loading phasors, with the solid component defining phase zero; amplitudes are fractional clock rates in units of 10^{-18}. **c,** Actual UTC sampling in eight observing periods, with accepted nominal member-hours. **d,** The S_2-M_2 astronomical phase at those same samples, including nodal corrections. Summer periods P6–P8 are orange. Geography and phasors describe the prediction; the lower panels describe the observation window.

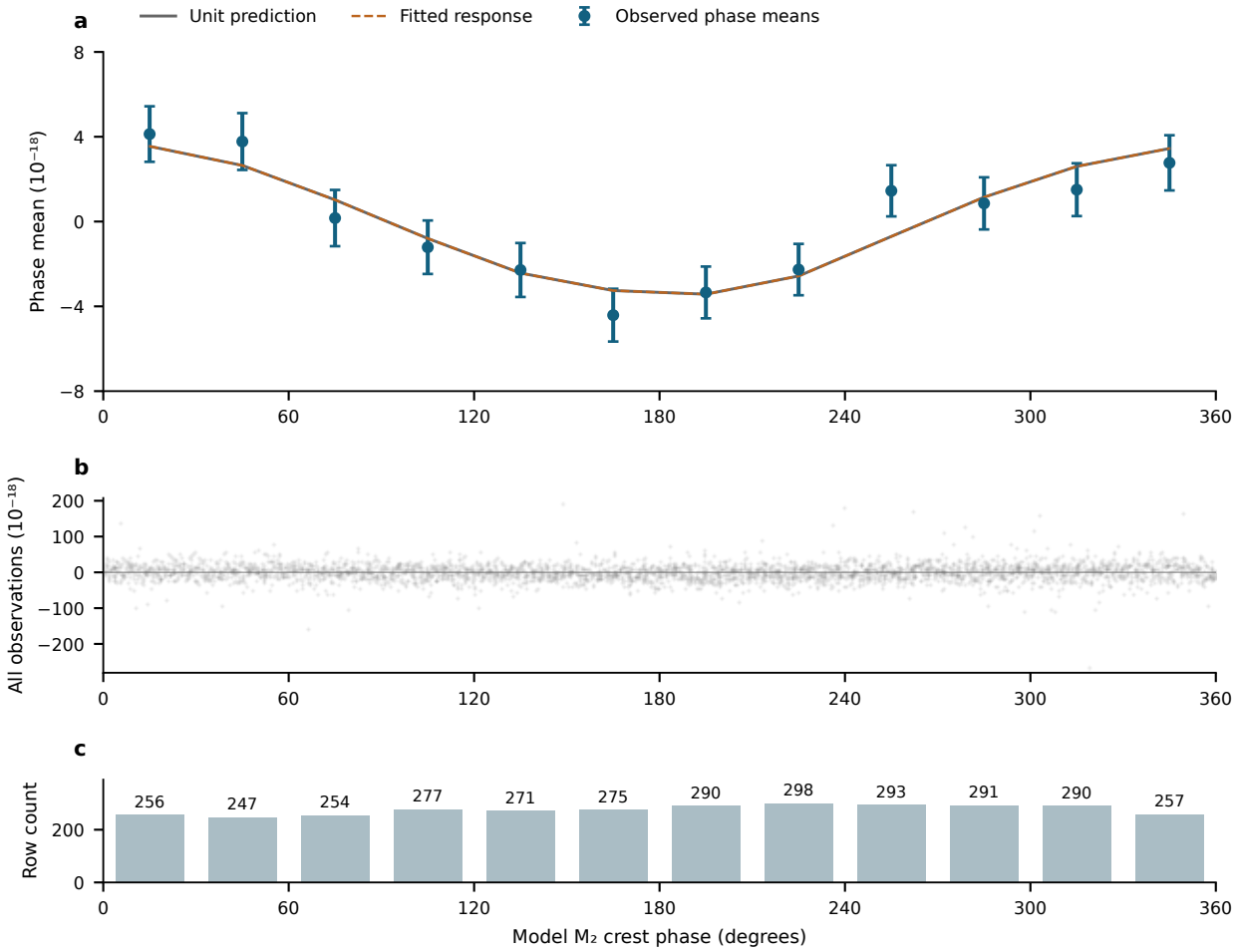


Fig. 2 | The full record ordered by lunar phase. a, Twelve 30° M_2 phase means of all 3,299 observations, after subtracting the fixed minor remainder and projecting against 54 segment constants and the neighbouring semidiurnal and daily groups. Identical weighted projection and averaging are applied to the unit lunar prediction; the fitted curve uses the unbinned primary coefficient. Error bars are one working standard error with the nuisance projection propagated; phase means are correlated. **b,** Every projected observation, with the full observed vertical range. **c,** Number of rows in each bin. Data, weights and templates are the primary segment design throughout. This is a display of the main observations, not an independent experiment.

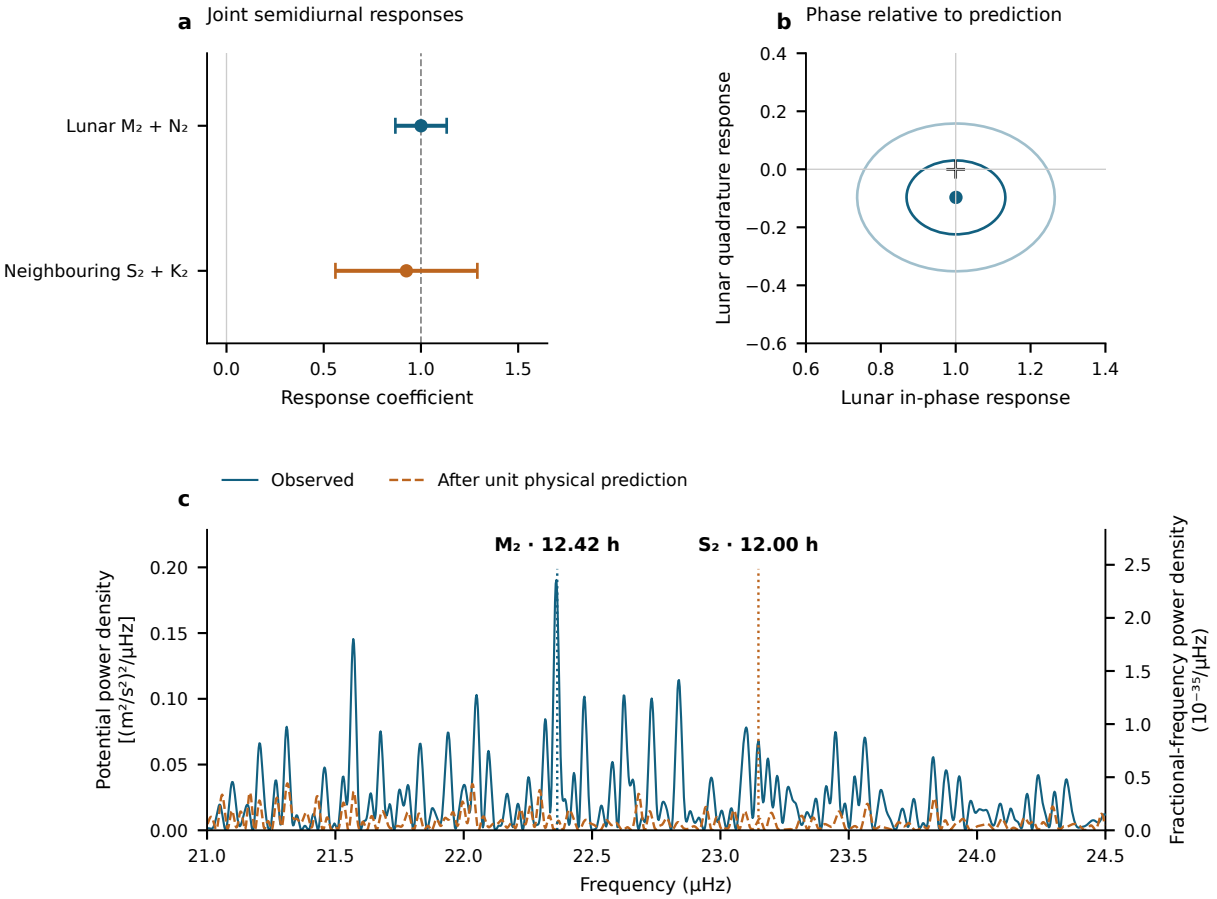


Fig. 3 | Semidiurnal responses, phase and measured spectral structure. a, The two semidiurnal coefficients from the three-group joint model, using all 3,299 means. Error bars are one standard error, defined as the larger of scaled least-squares and UTC-day jackknife errors. The daily group remains in the fit and is reported in Methods and Extended Data Table 2. **b,** Lunar in-phase and quadrature coefficients; the cross marks the unit prediction. Contours have covariance radii one and two from scaled least squares, not separately calibrated joint confidence levels. **c,** Equal-weight spectral calculation around the principal M_2 and S_2 frequencies, separated by 0.78 μHz: 3,299 real UTC means, 54 segment constants and equal weights. Blue shows the observed scan; orange follows subtraction of the fixed unit total prediction. No gap is filled or compressed. Both axes show the same nominal one-sided projection density per μHz, related by $S_W = c^4 S_y$. Sampling sidelobes contribute about 8% bidirectional M_2/S_2 cross-power. Peak locations support the principal-frequency attribution; group coefficients in **a** use the physical templates and Allan-based weights. All panels use the same observation rows.

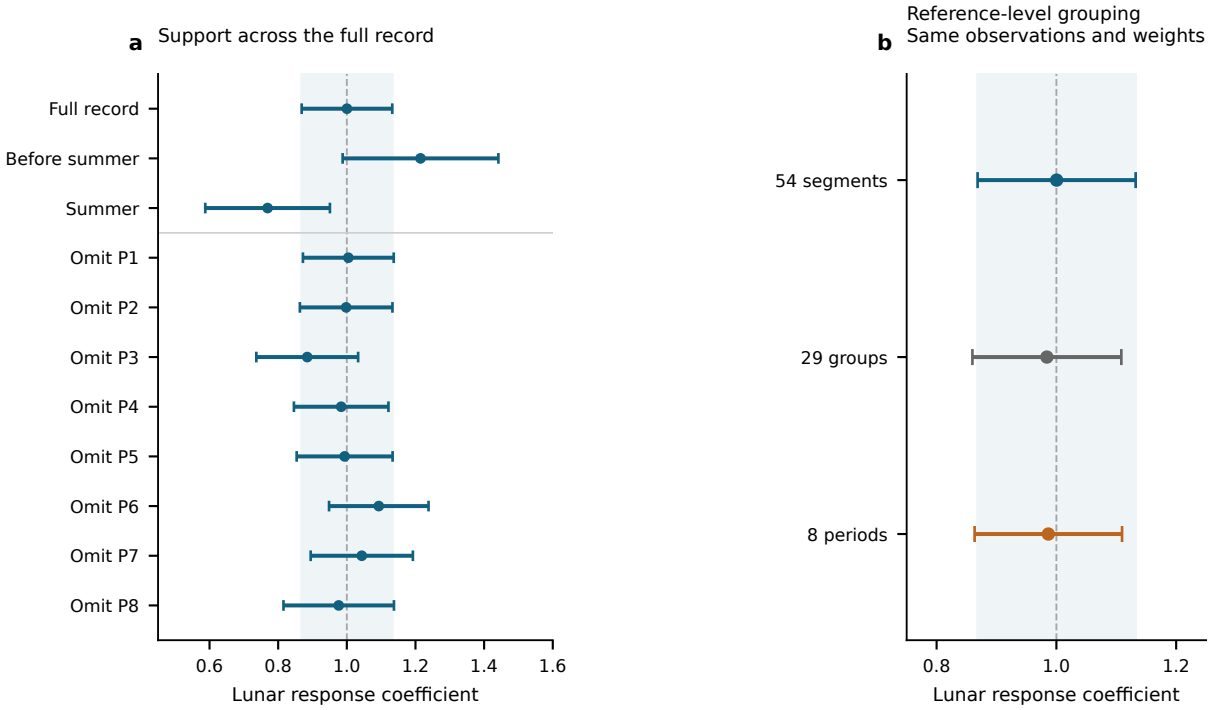


Fig. 4 | Consistency across observing periods and reference-level designs. a, Lunar estimates from the full record, the pre-summer and summer subsets, and deletion of each observing period. Each fit uses segment constants and the primary noise weights, with errors recalculated on its retained rows. The full, pre-summer and summer fits contain 3,299, 1,651 and 1,648 means. Deleting P1–P8 retains 3,136, 3,096, 2,890, 2,852, 2,870, 2,698, 2,704, 2,847 means, respectively. **b,** Lunar response with 54 segment, 29 operating-group or eight period constants. Only the intercept grouping changes; all designs use the same 3,299 observations, physical templates, corrections and noise weights. Error bars are one standard error using the same larger-of-scaled-least-squares-and-day-jackknife rule. Shading marks the full-record segment estimate and its standard error. The estimates share observations; reference-level choice follows the available calibration records.

Methods

Observations, frequency sign and averaging

The Shanghai and Wuhan coordinates used by the frozen prediction are 31.130° N, 121.570° E and 30.529° N, 114.357° E. The measured ratio is Wuhan Yb/Shanghai Sr. With positive geodetic potential W, its tidal fractional variation is $-[\delta W_{\mathrm{Wuhan}} - \delta W_{\mathrm{Shanghai}}]/c^2$. The period-relative beat conversion is $-d_p(f_{\mathrm{beat}} - f_{\mathrm{ref},p})/F_{1550}$, where $F_{1550} = 1.933992 \times 10^{14}$ Hz and $d_p = 2$ in 2025 and 1 in 2026. Thus a 2026 beat deviation of 1 Hz has fractional magnitude 5.17×10^{-15}. The 2025 division factor is applied before forming the small clock-rate variation.

The supplied valid-interval table defines 54 segments in 29 operating groups and eight periods. Counter members are read across midnight before binning. Beijing-time labels are converted to UTC. Summer group corrections follow the decimal-arithmetic frequency-chain calculation; early absolute correction entries remain incomplete. Their constant part is represented by segment intercepts, while any time variation within a segment remains a possible systematic contribution. Fitted intercepts are never applied to join the runs for long-term stability estimation.

The membership audit contains 1,966,908 nominal one-second rows before the minimum-bin-support cut. Three boundary rows reconcile its difference from the printed interval total. Means are formed in nominal 600-s bins separately within each segment and require at least 180 retained members. The 3,299 accepted means contain 1,965,327 members, or 545.924 h. There are 3,291 unique bin centres: eight centres contain separate partial means from different segments, with disjoint members. The 353.39-day span contains observations on 43 UTC dates. The inherited valid mask is retained; this work neither applies an additional triangular filter nor selects points using tidal residuals. Exact raw-file memberships and hashes are retained in the controlled archive.

What the constituent names mean

Semidiurnal tides produce approximately two cycles per day. M_2 is the principal lunar constituent; N_2 describes an additional lunar variation associated with orbital ellipticity. S_2 is the principal solar semidiurnal constituent, whereas K_2 contains lunar and solar forcing. The names identify astronomical frequencies, not separate instruments or independent experiments. The diurnal group contains K_1 (lunisolar), O_1 (principal lunar), P_1 (principal solar) and Q_1 (lunar elliptic). Their periods are listed in Extended Data Table 1.[22] Internal relative amplitudes, phases and nodal modulation are fixed within each physical group. A group response therefore tests that complete specified pattern.

Combined solid and loading potential

The physical target includes the direct astronomical potential, the Earth's induced potential and the effect of clock displacement. The selected frozen solid sequence implements these through the IERS radial displacement u and a fixed effective-potential conversion,

$$y_{\mathrm{solid}} = -\frac{1 + k_2 - h_2}{h_2} \frac{g(u_{\mathrm{Wuhan}} - u_{\mathrm{Shanghai}})}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794\,\mathrm{m/s^2}$. This is a displacement-based approximation to the combined solid potential, not the alternative frequency-resolved direct-potential implementation. The same conversion also acts on degree-three and frequency-dependent displacement contributions in the supplied sequence. A previous implementation comparison found an approximately 18% K_1 amplitude deficit relative to a potential-weighted reference. The daily response is accordingly not used to infer a Love number. Uncertainty in the solid model remains separate from the quoted statistical errors.

For loading, FES2022 sea-surface elevations ζ are convolved with the effective-potential kernel

$$\delta W_{\mathrm{load}} = \int \rho_w \zeta(\mathbf{r}', t) G_W(\psi)\, \mathrm{d}A, \qquad G_W(\psi) = \frac{ga}{M_E} \sum_n (1 + k'_n - h'_n) P_n(\cos\psi).$$

The three terms represent direct water attraction, induced potential and clock displacement. PREM load Love numbers are retained through degree 696 with the archived high-degree continuation. Eighteen loading constituents are synthesized with their astronomical nodal corrections. The complex convention is $f\,\mathrm{Re}[C\exp(-i\phi)]$; reversing the exponential sign requires conjugating C. The present analysis reads the frozen constituent and time-series tables rather than substituting an older loading sequence.

The eight named constituents form the three physical groups. Minor terms remain once as a fixed remainder r_0. On the actual weighted segment-centred sampling, solid, loading and total root-mean-square variations are respectively 2.928, 1.328 and 4.011×10^{-18}. Their ratio 0.331 is an amplitude-norm ratio, not an additive fraction of total power. The separately reported 0.89 ± 0.55 loading coefficient belongs to the alternative directly modelled, filtered reduction. It is unresolved and has not been recalculated as a separate coefficient of the present estimator.

Joint estimator and zero-point choice

We fit

$$y_i = c_{s(i)} + A_L s_{L,i} + A_S s_{S,i} + A_D s_{D,i} + r_{0,i} + \epsilon_i,$$

with 54 segment intercepts and three physical response coefficients. Weighted least squares uses the frozen Allan-based uncertainties σ_i, with $W_{ii} = 1/\sigma_i^2$. A period noise coefficient α_p is estimated from within-segment overlapping Allan deviations at 256, 512, 1,024, 2,048 and 4,096 s, after removing the fixed unit tide. No Allan window crosses a segment boundary. The bin uncertainty is $\sigma_i = \alpha_{p(i)}/\sqrt{n_i}$, where n_i is its retained member count. This is a working noise model, not stability

calculated after joining fit-adjusted levels. Independent QR replay reproduces the eight-period design on the same rows before changing only its intercept grouping. All comparisons therefore hold sampling, correction values, physical templates and weights fixed.

The daily group is an auxiliary physical component, fitted simultaneously to propagate its covariance with the semidiurnal responses. Its coefficient is $A_D = 0.58 \pm 0.54$ (Extended Data Table 2). Its predicted variation is smaller on this baseline, and its longer period supplies fewer cycles within short segments, increasing overlap with segment reference levels and slow variations. This auxiliary estimate is not a separate measurement objective.

The inspected interval and correction records contain no documented calibrated zero transfer for the 46 adjacent segment boundaries that share an eight-period intercept. They give valid times, but not a signed boundary transfer with uncertainty or a lock/relock history establishing unchanged reference levels. This motivates separate segment constants. The same-data residual and deletion diagnostics are reported in Extended Data Fig. 2 and Table 2. Coefficient proximity to one is not a model-selection rule.

Statistical errors and conditional coverage

For each group the reported standard error is the larger of residual-scaled least squares and leave-one-UTC-day-out jackknife errors. Deletions remove a calendar day, refit all remaining coefficients and drop empty intercept columns. The jackknife variance is $(G-1)G^{-1}\sum_g(\hat{A}_{(-g)} - \overline{A})^2$. Period-deletion errors are recorded separately. Extended Data Table 2 gives all components and the same-data intercept comparisons. Observations are repeated measurements of one clock pair, not thousands of independent experiments.

Conditional simulations apply both designs to the same 3,299-row sampling and heteroscedastic uncertainties. Each of four scenarios has 1,000 realizations: independent Gaussian noise; equal variance contributions from Gaussian white noise and

a 6-h Ornstein–Uhlenbeck process within segments; the corresponding 24-h process within periods; and standardized Student-t_5 noise. The overall scale is the observed segment-model residual scale; all true physical response coefficients are one. Each realization recomputes both components of the standard-error rule. Seed 2026110617 fixes the simulation. Coverage of $\hat{A} \pm 1.96u$ is 95.1–97.4% for the segment-model lunar coefficient and 95.4–96.8% for its neighbouring coefficient; Monte Carlo standard errors are about 0.4–0.7 percentage points (Extended Data Table 3). These are tests under declared noise laws, not a calibration against every possible disturbance.

The phase-relaxed model adds the lunar quadrature while retaining the prescribed M_2/N_2 relationship. Covariance contours use scaled least squares. Subset and deletion fits refit the same model on their retained rows. Shared-data estimates are not averaged. The simulation coverage refers to two-sided intervals; the reported $\pm$ values are standard errors.

Phase means, spectral density and sampling sidelobes

For the phase plot, N contains segment constants and the two non-lunar groups. We project both $y - r_0$ and the unit lunar prediction with $M_N = I - N(N^T W N)^{-1} N^T W$, and average in twelve fixed M_2 phase bins. If B is weighted bin averaging, the mean covariance is $B M_N C_y M_N^T B^T$, where C_y is the scaled working observation covariance. It retains correlations between means and propagates nuisance estimation. The full unbinned response remains primary.

Figure 3c preserves the independently specified equal-weight, 54-intercept scan of the unchanged observed means, rather than replacing it with the amplitude fit's weights. Every trial sine–cosine pair and the data are projected against the same constants. For a reduction in squared residuals $\Delta\mathrm{RSS}(f)$ with dimensionless y, the nominal one-sided projection density is $S_y(f) = 600\,\mathrm{s}\,\Delta\mathrm{RSS}(f)$ per Hz, or 10^{-6} times that value per µHz. Potential density is $S_W = c^4 S_y$; amplitude density is $c^2\sqrt{S_y}$. Uniform-grid checks reproduce the conventional one-sided Fourier density. For interrupted data this normalization is a stated projection density, not a Parseval-calibrated

variance over a filled year.[23]

The frequency step is 0.002729 µHz; the inverse calendar span is 0.032752 µHz. Fine plotting steps do not improve physical resolution. M_2-only and S_2-only astronomical injections use the same actual times, nodal factors and equal-weight segment projection. Power at the opposite principal frequency, divided by power at the injected frequency, gives 0.08027 and 0.08149. Conditional scans project out a freely phased sinusoid at the other frequency before scanning the target neighbourhood. Amplitude interference means that such corrections act on sampled waveforms or complex coefficients, not by subtracting scalar power spectra. Gaps alone do not predict the realization of measurement noise. Residual density can describe remaining measurement fluctuations, but includes possible model mismatch and cannot be called a pure instrumental noise floor.

Instrument and daily-response checks

The matched monitor calculation uses 1,482 rows with 887,882 nominal members, or 246.634 h, in P6–P8. Both baseline and augmented fits have identical members, segment labels and weights. Paired day-deletion replicates propagate uncertainty in their difference. Extended Data Fig. 3 reports the response changes and dates. This comparison concerns the monitored transmission route during the measured overlap.

On the selected data and model, a common 24-h sine–cosine pair gives $A_L = 1.036 \pm 0.133$; separate 24-h pairs in each period give 1.003 ± 0.136. These daily sensitivity designs address a periodicity motivated by temperature and operating schedules, without assigning an independently measured thermal amplitude. Free 12-h pairs in each period give 0.751 ± 0.242 and a very poorly constrained neighbouring coefficient because they compete with astronomical semidiurnal variation within short periods. That fit is retained as a sensitivity boundary.

The archived single-bin diagnostic and separately filtered reduction are documented in Extended Data Table 4. The latter uses different filtering, weighting, a direct-

potential solid model and a residual-shift uncertainty; its manuscript-reported results are distinguished from the present replay. Its ERA5 air-temperature regressors are environmental proxies, distinct from the stated chamber-thermometer record.

AI-assisted preparation

OpenAI Codex assisted English drafting, numerical replay and figure preparation. [Authors to confirm their verification of the analyses, citations and final text, and responsibility for the submitted work.]

Data availability

The author review package contains the 3,299 derived means, fixed templates, noise weights, figure source tables and immutable input hashes. The original counters and membership ledger are held in the controlled experimental archive. [Authors to supply an editor-accessible location and confirm the permanent data-access statement and repository identifier before submission.]

Code availability

The package contains the independent QR replay, same-data design comparison, conditional simulations, spectral calculations, covariance propagation and figure scripts, with software versions and a reproducible entry point. [Authors to confirm the code licence and permanent repository identifier before submission.]

Methods references

22. NOAA. Harmonic constituents. *NOAA Tides and Currents*. https://tidesandcurrents.noaa.gov/harcon.html?id=8577330 (accessed 6 November 2026).
23. VanderPlas, J. T. Understanding the Lomb–Scargle periodogram. *Astrophys. J. Suppl. Ser.* **236**, 16 (2018). https://doi.org/10.3847/1538-4365/aab766

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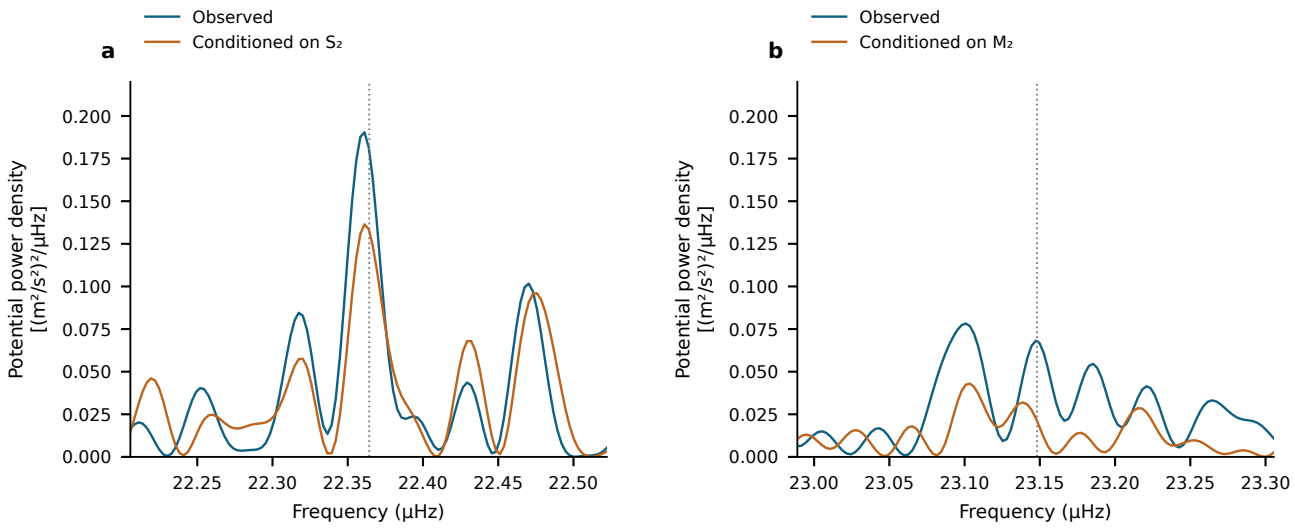
Competing interests

[Declaration to be supplied by the authors.]

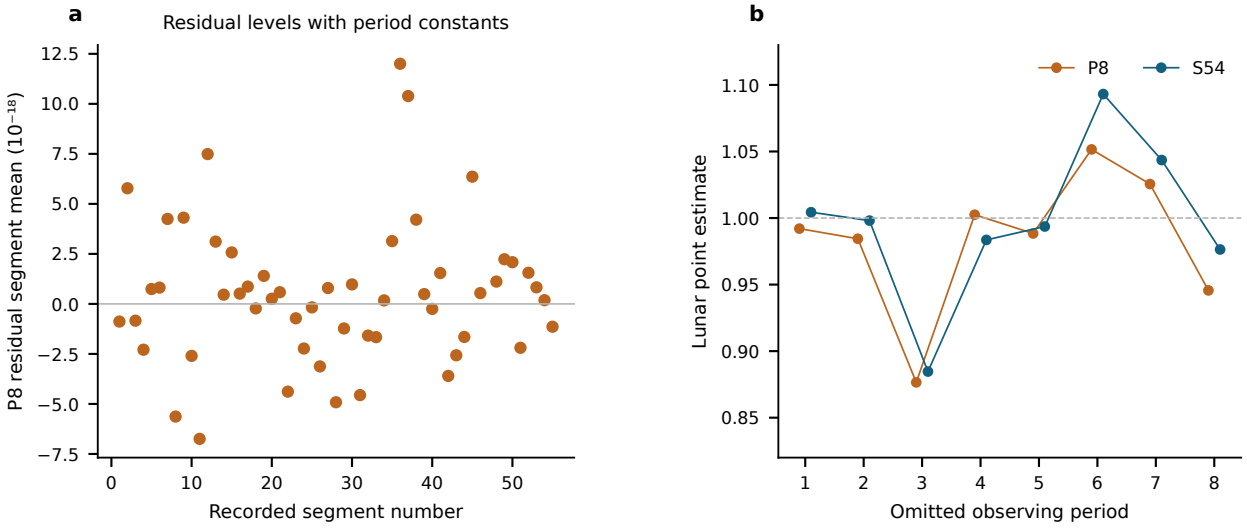
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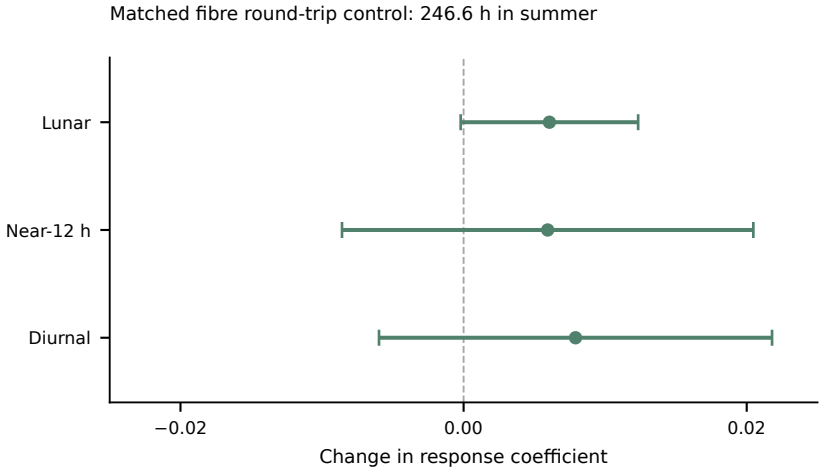
Extended Data



Extended Data Fig. 1 | Conditional semidiurnal scans. Equal-weight scans of the same 3,299 observations with 54 segment intercepts. **a,** M_2 neighbourhood before and after projection of a sine–cosine pair at S_2. **b,** S_2 neighbourhood before and after projection at M_2. Dotted lines show the astronomical target frequencies. These are descriptive projections on shared observations, not additional detection probabilities.



Extended Data Fig. 2 | Same-data comparison of reference-level designs. a, Weighted segment-mean residuals after the eight-period joint fit, ranging from -6.75 to 12.00×10^{-18}. Segment constants make these means zero by construction. **b,** Lunar point estimates after period deletion, spanning 0.877–1.052 with eight period constants and 0.885–1.093 with 54 segment constants. The weighted residual sums of squares are 3,781.58 and 3,713.64; the latter has 46 additional intercept parameters. Unweighted residual root-mean-square values are 24.114 and 23.934×10^{-18}, and reduced residual sums of squares are 1.1501 and 1.1455. Adjacent standardized residual correlations are −0.127 and −0.148, using pairs within the same segment and separated by at most 900 s. All comparisons retain identical observations, templates, corrections and noise weights. These are residual diagnostics, not measurements of physical jumps or zero-transfer calibration.



Extended Data Fig. 3 | Simultaneous summer transmission control. Changes caused by adding the fibre round-trip monitor to the joint model, with paired UTC-day-jackknife standard errors. Baseline and augmented fits use identical clock/monitor members, labels and weights: 1,482 means containing 887,882 nominal one-second members, or 246.634 h in P6–P8, from 29 June to 25 August 2026. The lunar, neighbouring and daily shifts are respectively 0.00606 ± 0.00627, 0.00594 ± 0.01453 and 0.00791 ± 0.01388. The paired monitor constrains the measured transmission route on this summer overlap. The main tidal estimate uses the entire 545.9-h clock record. The prior transmission-architecture demonstration over 2,067 km is a separate experiment.[16]

Extended Data Table 1 | Astronomical constituent guide. Names specify forcing and frequency; one response coefficient multiplies each complete group with its internal ratios and phases fixed.

Symbol	Period_h	Meaning	Group
M2	12.421	Principal lunar semidiurnal	Lunar
N2	12.658	Lunar elliptic semidiurnal	Lunar
S2	12	Principal solar semidiurnal	Near-12 h
K2	11.967	Lunisolar semidiurnal	Near-12 h
K1	23.934	Lunisolar diurnal	Diurnal
O1	25.819	Principal lunar diurnal	Diurnal
P1	24.066	Principal solar diurnal	Diurnal
Q1	26.868	Lunar elliptic diurnal	Diurnal

Extended Data Table 2 | Estimator and uncertainty comparison. All directly replayed designs use the same 3,299 means, physical templates and Allan-based weights. Errors are standard errors. The alternative filtered estimate is separately labelled author-reported.

Design	Lunar	Near_12h	Diurnal	Status
S54	1.000 ± 0.132	0.925 ± 0.365	0.579 ± 0.538	Replayed
G29	0.984 ± 0.124	1.093 ± 0.298	0.759 ± 0.420	Replayed
P8	0.987 ± 0.123	1.129 ± 0.288	0.678 ± 0.359	Replayed
Filtered, 54 segments	0.97 ± 0.14	0.90 ± 0.29	0.35 ± 0.51	Author-reported

Group	Estimate	Scaled_LS	Day_JK	Period_JK	Reported_SE
Lunar	1.0004	0.132	0.1218	0.147	0.132
Near-12 h	0.9251	0.2993	0.3649	0.3531	0.3649
Diurnal	0.5792	0.3887	0.5381	0.7735	0.5381

Extended Data Table 3 | Conditional 95% interval coverage. Each scenario has 1,000 simulated realizations per design, with fixed observed sampling and weights. Values are percentages for the larger-of-LS-and-day-jackknife rule, using multiplier 1.96. They assess the specified scenarios, not unknown systematic effects.

scenario	design	Diurnal	Lunar	Near12h
50% OU 24 h, period	P8	97.9	98	97.3
50% OU 24 h, period	S54	96.4	95.1	95.4
50% OU 6 h, segment	P8	94.7	95.7	96
50% OU 6 h, segment	S54	95.8	95.6	96.3
Student t (df 5)	P8	96.8	96.3	96.2
Student t (df 5)	S54	96.2	96.1	96.8
Gaussian	P8	95.7	96.2	95.9
Gaussian	S54	95.5	97.4	96.7

Extended Data Table 4 | Archived influence diagnostic and alternative reduction. These rows identify the observation support and estimator of each result. The single-bin diagnostic retains its original eight-period design. The filtered reduction is reported in the alternative manuscript (v68); its measurements overlap the present data and its results are not pooled with the primary estimates.

Check	Observation support and design	Result and status
Pre-identified extreme mean	2026-07-07 01:25 Beijing time, segment 46; archived eight-period model, fixed templates and weights	Omitting the mean changes the daily coefficient from 0.678 to 0.721: +0.043 versus its original 0.359 standard error. The lunar change is +0.039. Uncertainty was not recalibrated. The mean remains in the present 3,299 rows.
Filtered lunar response	Approximately 544 h, 54 segments; 1,200-s triangular filter every 600 s; equal weights, direct-potential solid model; continuous-run circular-residual-shift error	0.97 ± 0.14, author-reported. The other grouped results are in Extended Data Table 2.
Separate loading coefficient	Same alternative filtered reduction; loading assigned a separate coefficient	0.89 ± 0.55, author-reported and unresolved. This is not a separate loading estimate from the primary design.
Temperature regressors	Two-site ERA5 2-m air-temperature proxies in the alternative reduction; distinct from its stated 98-h Sr chamber-thermometer record	Lunar 0.98 ± 0.13, author-reported. Exact regression membership is needed for independent replay.
Alternative round-trip control	Reported 474-h overlap in the filtered reduction	Lunar and whole-template coefficient changes less than 0.005, author-reported. Exact matching memberships are needed for independent replay; the directly replayed 246.634-h control is shown in Extended Data Fig. 3.

The archived influence diagnostic is retained for provenance; the current observation membership is unchanged.